

Industry's lowest active current for 16-bit microcontrollers

Flexible power-saving features plus on-chip LCD driver

eXtreme Low Power 16-bit MCUs

Active Current 150 μ A/MHz
RAM Retention 330 nA
VBAT Battery Backup With RTCC 400 nA

A new industry lowest active current, automatic transition from V_{DD} to V_{BAT} battery backup, and low-power sleep with RAM retention combine to help you to push battery life towards the end of your product's operational lifetime.

The combination of innovative and flexible power-saving features, an on-chip LCD driver and mTouch™ capacitive sensing on Microchip's new PIC24F 'GA3' family lets you create more power efficient and less costly battery-backed products.

As the first PIC® microcontrollers with V_{BAT} , the PIC24F 'GA3' family maintains system time by automatically switching to back-up battery power when the primary V_{DD} supply is removed. This extends battery life by driving the application's real-time clock whilst drawing only 400 nA.

An industry lowest 150 μ A/MHz active current for 16-bit microcontrollers is combined with a new low-power sleep mode with RAM retention down to 330 nA, and six DMA channels to cut power consumption and increase the throughput when executing routines.

Adding more informative and flexible displays is easy with the on-chip LCD display driver; whilst mTouch™ capacitive sensing, ultrasonic flow measurement and other sensor functions can be added using the on-chip Charge Time Measurement Unit (CTMU) with constant current source.

3 EASY STEPS TO ENERGY EFFICIENT DESIGNS:

1. Use new power-saving features to extend battery life
2. Add mTouch™ capacitive sensing, flow measurement or other sensor functions
3. Use the LCD driver to directly drive a large segmented display
www.microchip.com/16bit

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